BENGALURU SMART MOBILITY INTELLIGENCE

A data-driven platform for traffic forecasting, road safety analysis, enforcement insights and camera-based vehicle intelligence

An integrated machine-learning and computer-vision decision-support prototype for urban mobility in Bengaluru.

1. Executive Summary

Bengaluru Smart Mobility Intelligence is an applied data science project designed to support traffic planning by combining historical traffic observations, road-safety records, traffic-police enforcement statistics, mobility indicators and camera-based vehicle detection in one interactive application. The system provides road-level congestion estimates, next-day forecasts, severe-congestion risk indicators, descriptive analysis of crashes and enforcement, and a camera-derived visual congestion score.

The project uses Python for data preparation and modelling, machine-learning methods for prediction, YOLO-based vehicle detection for image/video analysis, and Streamlit with interactive Plotly visualisations for delivery. The intended outcome is a decision-support prototype that helps users explore traffic patterns and identify locations that may need closer attention. Predictions and priority scores are advisory, not guarantees or automated instructions to traffic authorities.

2. Problem Statement

Background

Urban traffic congestion affects travel time, road capacity, mobility and the ability of road users to move safely. Traffic, crash and enforcement information can be stored in separate datasets, making it difficult to review these signals together or compare locations consistently.

Problem

A practical, unified tool is needed to summarise historical conditions, estimate congestion under user-selected conditions, provide a cautious next-day outlook, and present road-safety, enforcement and camera observations alongside the traffic context.

Objectives

- Prepare and analyse traffic, road-safety, enforcement and mobility datasets relevant to Bengaluru.
- Estimate congestion for a selected road using user-adjustable conditions such as traffic volume, incidents, rainfall, roadwork and mobility indicators.
- Forecast next-day congestion and estimate the probability of a severe congestion level.
- Analyse historical crash/fatality and traffic-enforcement patterns, including camera-based enforcement share where available.
- Use YOLO vehicle detections and passenger-car-unit (PCU) weights to calculate a camera-based load and visual congestion index.
- Combine model outputs and contextual indicators into an explainable, interactive dashboard for exploratory planning.

Scope

The current application covers 16 monitored roads across Bengaluru areas, with traffic records spanning approximately two and a half years in the dashboard dataset. Separate road-safety and enforcement tables are analysed over the years available in those sources. Camera analysis supports sample images/video, user uploads and a live-stream URL when a compatible and accessible stream is supplied. A selected road is a contextual association; it does not prove that a test stream physically originates from that road.

3. Data Sources

The project uses secondary datasets and project-derived features. Add exact dataset titles, publishers, URLs, licences and access dates if required for formal submission.

Dataset / input	Use in the project	Important caveat
Traffic records	Road-level volume, speed, congestion, travel-time index, capacity and related conditions.	Coverage is limited to monitored roads and available dates.
Weather and road conditions	Weather, rainfall and roadwork indicators used for analysis and prediction.	Associations are not proof of causation.
Road-crash statistics	Yearly crashes, fatal crashes, deaths and derived safety ratios.	Latest year may be incomplete; forecasts are exploratory.
Traffic-police enforcement records	Cases booked, fines, enforcement categories and automation share.	Inconsistencies and reporting changes may affect comparisons.

Dataset / input	Use in the project	Important caveat
BMD-45 vehicle imagery	Training/validation examples for 14-class YOLO vehicle detection.	Performance depends on image quality and viewpoint.
Camera image, video or RTSP input	Detect vehicles and estimate PCU load / visual congestion.	Test footage must not be labelled as actual local CCTV.

4. Methodology

4.1 Data preparation

Inspect data types; standardise dates and road names; check duplicates, missing values, outliers and inconsistent totals; derive calendar, lag and rolling features where appropriate; and preserve chronology for time-based evaluation.

4.2 Exploratory and descriptive analysis

Summarise congestion by road, area, date and weekday; compare it with capacity utilisation, incidents, weather, roadwork and public-transport indicators; and visualise crash and enforcement trends.

4.3 Congestion estimation and forecasting

Use the trained machine-learning pipeline to estimate congestion for user-selected conditions and forecast the next day. Evaluate using chronological holdout data and MAE, RMSE and R^2, alongside simple baselines.

4.4 Incident-risk analysis

Estimate expected incident counts from available traffic, weather, roadwork, calendar and road features. Treat outputs as prioritisation indicators, not exact incident or casualty predictions.

4.5 Camera-based vehicle intelligence

Run YOLO on supplied frames, classify vehicles, apply configured PCU factors and calculate a relative visual congestion index. Associate camera results with the selected road's model-derived context for a planning-priority summary.

4.6 Dashboard and application

Present prediction, city map, camera view, traffic overview, road safety, enforcement, mobility and model insights in Streamlit with interactive charts and tables.

Evaluation approach

- Use time-ordered training and test periods to reduce look-ahead leakage.
- Report held-out error metrics; R^2 is not the same as accuracy.
- Compare models with simple baselines and account for day-to-day traffic noise.
- Report vehicle-detection mAP and PCU-load agreement on held-out frames where available.
- Test valid files and accessible streams; separate stream access failures from model performance.

5. Expected Outcomes

- A Streamlit prototype combining traffic, safety, enforcement and mobility insights.
- Road-level congestion estimates and next-day outlooks with cautious interpretation.
- Dashboards highlighting traffic pressure, congestion patterns, crash trends and enforcement patterns.
- A YOLO camera workflow reporting vehicle detections, estimated PCU load and relative visual congestion.
- A transparent prioritisation summary for further investigation.
- Documented evaluation, limitations and recommendations for future improvement.

6. Risks and Challenges

Risk / limitation	Mitigation / handling
Incomplete or duplicated records	Profile data quality, validate totals and document corrections.
Limited geographic/time coverage	Avoid generalising results to every Bengaluru road.
Forecast uncertainty	Use chronological testing, baselines and clear prediction caveats.
Correlation mistaken for causation	Use association-focused language and avoid unsupported causal claims.
Detector errors	Evaluate held-out frames and note lighting, angle and occlusion limits.
Unavailable/authenticated camera stream	Use authorised streams and label test footage clearly.
Privacy and responsible use	Protect stream credentials and avoid identifying people.

7. Deliverables and Indicative Work Plan

Stage	Work	Deliverable
1. Data review	Inspect sources, clean data and document coverage.	Prepared data and quality notes.
2. Analysis	Explore traffic, safety, enforcement and mobility.	Charts and descriptive findings.
3. Modelling	Train/evaluate forecasting, risk and vision components.	Models and evaluation results.
4. Integration	Connect models and data to Streamlit.	Interactive application and usage notes.
5. Validation	Test inputs, outputs, errors and limitations.	Test checklist and final report.

The stages are indicative and can be mapped to the required academic calendar.

8. Conclusion

Bengaluru Smart Mobility Intelligence brings together predictive modelling, descriptive analysis and computer vision in a single decision-support prototype. Its value lies in presenting multiple sources of traffic-related evidence in a form that users can explore, compare and use to identify locations for further review. It does not replace professional traffic engineering or official emergency and enforcement processes. Findings must be interpreted in light of data limitations, model uncertainty and the distinction between correlation and causation.

References and source documentation

Add formal references for the traffic, crash, enforcement and BMD-45 datasets, plus official documentation for Python, scikit-learn, Streamlit, Plotly, Ultralytics/YOLO and OpenCV used in the implementation.

End of proposal